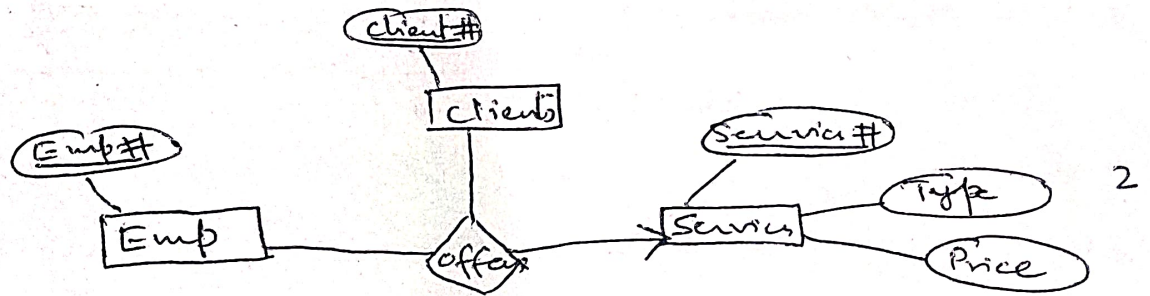


- Primary Key of Branch is Branch # $\frac{1}{4}$
- Primary Key of Services is Service # $\frac{1}{4}$
- ~~PK~~ Primary Key of Offers is (Branch #, Services #) $\frac{1}{2}$

Cardinality $\rightarrow$ Many-to-Many (M:N) $\frac{1}{1}$

Participation Constraints: Partial on both sides $\frac{1}{1}$



Primary Key (PK) of Emp is Emp #

" " client is client #

" " Services is Service #

" " Offers is (Emp #, Client #) $\frac{1}{2}$

PK $\frac{1}{2}$

Cardinality [Emp: client: Services] :: (N:N:1) $\frac{1}{1}$

Participation Constraints: Partial on all sides $\frac{1}{1}$

③ There is NO weak entity, because $\frac{1}{1}$

- All entities [Emp, Branch, Client, Services] have their own Primary Key.
- These entities are not dependent on any entity $\frac{1}{1}$
- No entity has a partial Key.

PKs are Underlined

④ ~~Branch (B#, Name, ...)~~

Employee (E#, Name, ..., S#), where S# is FK

① Service (S#, Type, Price, ...)

Client (C#, Name, ...)

PK } ①
FK } ②

~~Manages (B#, E#, S#), where B# is Unique~~

(E#, S#) (S# → FK)

~~Provides (B#, S#), where B#, S# each FK~~

⑥ ~~Service Transaction (E#, C#, S#, ServiceDate)~~ PK is (E#, C#)
PK E#, C# S# } are FK
C# }
② marks each in Q 5 & 6

⑤ / A

⑥ RAE: $\gamma_{ServiceDate, COUNT(ClientID) \rightarrow Total clients (Provides)}$

SQL: Select ServiceDate, Count(ClientID) AS TotalClients
From Provides
Group By ServiceDate;

③

RAE $\pi_{ClientID, Name}(\sigma_{City='Kalkaji'}(Client))$

SQL: Select ClientID, Name
From Client
Where City = 'Kalkaji';

⑦

$P_1 = P(Provides)$ $P_2 = P(Provides)$

RAE = $\pi_{P_1, ClientID}(\{P_1, ClientID = P_2, ClientID\} (P_1 \bowtie P_2))$

SQL: Select ClientID From Provides
Group by ClientID Having COUNT(S#) ≥ 2; $\{P_1, S\# \neq P_2, S\# \}$ Join condition
SQL: Select Distinct P1.ClientID From Provides P1, Provides P2
AND P1.S# ≠ P2.S#

⑤

$P_1 = P$ (Provides)

$P_2 = P$ (Provides)

RAE: $\Pi_{P_1 \text{ clientID}} (P_1 \bowtie P_2)$
 $P_1 \text{ clientID} = P_2 \text{ clientID}$
 $\wedge$
 $P_1 \text{ EmpID} = P_2 \text{ EmpID}$
 $\wedge$
 $(P_1 \text{ S\#} \neq P_2 \text{ S\#})$

SQL: Select Distinct $P_1 \text{ clientID}$
 From Provides P_1 , Provides P_2
 Where $P_1 \text{ clientID} = P_2 \text{ clientID}$ AND $P_1 \text{ EmpID} = P_2 \text{ EmpID}$
 AND $P_1 \text{ S\#} \neq P_2 \text{ S\#}$

OR
 Select Distinct clientID
 From Provides
 Group By clientID , EmpID
 Having $\text{COUNT}(\text{S\#}) > 1$;

⑥

RAE $T = \Pi_{\text{S\#}} (\sigma_{\text{city} = \text{'Kalkaji'}} (\text{Branch}))$

Result = $\Pi_{\text{clientID}, \text{S\#}} (\text{Provides}) \div T$

SQL: Select Distinct clientID
 From Client C
 Where NOT EXISTS (
 (Select $\text{S\#}$ From Branch where $\text{city} = \text{'Kalkaji'}$)
 EXCEPT
 (Select $\text{P.S\#}$
 From Provides P
 Where $\text{C.clientID} = \text{P.clientID}$))

⑦

RAE

$E_1 = P(\text{Emp})$ $E_2 = P(\text{Emp})$
 $\Pi_{\text{city}} (\text{Emp}) - \Pi_{E_1 \text{ city}} (E_1 \bowtie E_2)$
 $E_1 \text{ city} = E_2 \text{ city} \wedge E_1 \text{ EmpID} \neq E_2 \text{ EmpID}$

SQL

Select City From Employee Group By City Having $\text{COUNT}(\text{EmpID}) = 1$;

②

④ RAE: $\Pi_{EmpID}(Employee) - \Pi_{EmpID}(Provides)$

SQL: (Select EmpID From Employee)

EXCEPT

(Select Distinct EmpID From Provides)

7

(a) Candidate ~~Primary~~ Key : Service transaction ID ✓

② ③

(b) { Transaction (Transaction ID, Emp ID, Branch ID, Service)
Employee (Emp ID, Emp Name) PK: ①
② Branch (Branch ID, Branch City)
Service (Service ID, Service Price, Service Duration)

Employee (EmpID, EmpName)

PK: ①

② Branch ~~III~~ (Branch II, Branch City)

Service (ServiceID, ServicePrice, ServiceDuration)

Highest Normal Form: BCNF



⑧ 1. Select Count(*) from client

2

2. Select ~~any~~ ~~attributes~~ ~~(of client notation)~~
From client where client ID = '.....';
 $\begin{matrix} > \\ < \end{matrix}$

2

From client where client ID = '.....';
 $\begin{matrix} > \\ < \end{matrix}$

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⑨ Create View Mumbai-Branch AS

②

Select * from Branch where City = 'Mumbai';

⑩ Select ServiceDate, count(*) AS Total Services
From Provides

②

From Provides

Group By Service Date

Order By Service Date;